Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Medical · Class 12

Subject: Biology

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: The Human Heart: Structure and Double Circulation

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Unit 1: The Human Heart: Structure and Double Circulation

Learning Objectives

• Students will trace the path of blood through the four chambers of the heart.

• Students will explain double circulation and why it keeps oxygenated blood separate.

• Students will state the function of valves in the heart.

Key Concepts

• The heart has four chambers: two atria on top and two ventricles below.

• The right side pumps blood to the lungs; the left side pumps blood to the whole body.

• Valves stop blood flowing backwards between chambers and into the great vessels.

• Double circulation means blood passes through the heart twice in one complete circuit.

Teaching Methodology (Step by Step)

1. Ask students where they feel their heartbeat and why it never stops.
2. Draw the heart on the board chamber by chamber, adding vessels as the story grows.
3. Trace deoxygenated blood from the body to the lungs, then oxygenated blood back to the body.
4. Compare the thickness of the two ventricle walls and link it to the distance blood must travel.
5. Students complete the double-circulation ordering activity in pairs.
6. Close by asking why a single circulation would mix oxygen-rich and oxygen-poor blood.

Classroom Activities

• Heart model walk: students trace the blood path through a labelled heart diagram.

• Double circulation race: two teams sort steps into pulmonary and systemic order.

• Pulse check: students find their own pulse and explain which side of the heart drives it.

Assessment Questions

1. Name the four chambers of the human heart in order of blood flow.
2. Why is the left ventricle wall thicker than the right ventricle wall?
3. Explain in two lines what double circulation means.

Homework

• Draw and label the human heart, marking oxygenated and deoxygenated blood in different colours.

• Write three differences between pulmonary and systemic circulation.

Time Allocation

40 minutes